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Department of Electronics Engineering

Course Name:	Object Oriented Programming	Semester:	III
Date of Performance:	___ / ___ / ____	Batch No:	Batch __A4__
Faculty Name:	Tejashree Phatak	Roll No:	16015024082
Faculty Sign & Date:		Grade/Marks:	___/15

Experiment No: 2

Title: First Java code use of Class and Object

Aim and Objective of the Experiment:

Learn the concept of encapsulation in Java

COs to be achieved:

CO1: Understand concepts of Object Oriented Programming and basic characteristics of Java.

Tools used:

Java v8,



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Theory:

1. (About all modifier used while declaring methods)

Modifiers are used to help us where the initialised method can be used.

i.e : static, private, public

static(only one instance is stored in memory and then used)

public - can be accessed anywhere, even outside of the class,,

Private - cannot be accessed outside the class.

2. (constructors vs methods)

Constructors are used to define the objects or to initialise the values in the objects.while methods are used to perform the logic in the class.

Code:

1. Write a program to perform addition, subtraction, multiplication and division of two complex numbers.
 - Variation 1: Implementation with One class only
 - Variation 2: Implementation with Two classes (One will be main class and Another will be class for complex) without constructors
 - Variation 3: variation2 with use of constructors (But without this keyword)
 - Variation 4: variation2 with use of constructors (But with use of this keyword)
 - Variation 5: Add one more class which performs following mathematical operation on complex number.
 - Inverse
 - Complex conjugate
 - Polar form

NOTE: use the concept of parameterized method, non-parameterized method, different constructors (passing all arguments, passing few arguments, passing object and without passing arguments)



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Output:

Var 1



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```
public class ComplexNos {
    double real, imag;

    ComplexNos() {
        real = 0;
        imag = 0;
    }

    ComplexNos(double r, double i) {
        real = r;
        imag = i;
    }

    ComplexNos add(ComplexNos c) {
        return new ComplexNos(real + c.real, imag + c.imag);
    }

    ComplexNos subtract(ComplexNos c) {
        return new ComplexNos(real - c.real, imag - c.imag);
    }

    ComplexNos multiply(ComplexNos c) {
        double r = real * c.real - imag * c.imag;
        double i = real * c.imag + imag * c.real;
        return new ComplexNos(r, i);
    }

    ComplexNos divide(ComplexNos c) {
        double denom = c.real * c.real + c.imag * c.imag;
        double r = (real * c.real + imag * c.imag) / denom;
        double i = (imag * c.real - real * c.imag) / denom;
        return new ComplexNos(r, i);
    }

    void display() {
        System.out.println(real + (imag >= 0 ? " + " : " - ") + Math.abs(imag) + "i");
    }

    public static void main(String[] args) {
        ComplexNos c1 = new ComplexNos(3, 2);
        ComplexNos c2 = new ComplexNos(1, 7);

        c1.display();
        c2.display();

        c1.add(c2).display();
        c1.subtract(c2).display();
        c1.multiply(c2).display();
        c1.divide(c2).display();
    }
}
```

Var 2 :



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```
class Complex {
    double real, imag;

    void setValues(double r, double i) {
        real = r;
        imag = i;
    }

    Complex add(Complex c) {
        Complex temp = new Complex();
        temp.real = real + c.real;
        temp.imag = imag + c.imag;
        return temp;
    }

    Complex subtract(Complex c) {
        Complex temp = new Complex();
        temp.real = real - c.real;
        temp.imag = imag - c.imag;
        return temp;
    }

    Complex multiply(Complex c) {
        Complex temp = new Complex();
        temp.real = real * c.real - imag * c.imag;
        temp.imag = real * c.imag + imag * c.real;
        return temp;
    }

    Complex divide(Complex c) {
        Complex temp = new Complex();
        double denom = c.real * c.real + c.imag * c.imag;
        temp.real = (real * c.real + imag * c.imag) / denom;
        temp.imag = (imag * c.real - real * c.imag) / denom;
        return temp;
    }

    void display() {
        System.out.println(real + (imag >= 0 ? " + " : " - ") + Math.abs(imag) + "i");
    }
}

public class ComplexNos {
    public static void main(String[] args) {
        Complex c1 = new Complex();
        Complex c2 = new Complex();

        c1.setValues(3, 2);
        c2.setValues(1, 7);

        c1.display();
        c2.display();

        c1.add(c2).display();
        c1.subtract(c2).display();
        c1.multiply(c2).display();
        c1.divide(c2).display();
    }
}
```

Var 3 :



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```
class Complex {
    double real, imag;

    Complex(double r, double i) {
        real = r;
        imag = i;
    }

    Complex add(Complex c) {
        return new Complex(real + c.real, imag + c.imag);
    }

    Complex subtract(Complex c) {
        return new Complex(real - c.real, imag - c.imag);
    }

    Complex multiply(Complex c) {
        double r = real * c.real - imag * c.imag;
        double i = real * c.imag + imag * c.real;
        return new Complex(r, i);
    }

    Complex divide(Complex c) {
        double denom = c.real * c.real + c.imag * c.imag;
        double r = (real * c.real - imag * c.imag) / denom;
        double i = (imag * c.real - real * c.imag) / denom;
        return new Complex(r, i);
    }

    void display() {
        System.out.println(real + (imag >= 0 ? " + " : " - ") + Math.abs(imag) + "i");
    }
}

public class ComplexNos {
    public static void main(String[] args) {
        Complex c1 = new Complex(3, 2);
        Complex c2 = new Complex(1, 7);

        c1.display();
        c2.display();

        c1.add(c2).display();
        c1.subtract(c2).display();
        c1.multiply(c2).display();
        c1.divide(c2).display();
    }
}
```

Var 4 :



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```
class Complex {
    double real, imag;

    Complex(double real, double imag) {
        this.real = real;
        this.imag = imag;
    }

    Complex add(Complex c) {
        return new Complex(this.real + c.real, this.imag + c.imag);
    }

    Complex subtract(Complex c) {
        return new Complex(this.real - c.real, this.imag - c.imag);
    }

    Complex multiply(Complex c) {
        double r = this.real * c.real - this.imag * c.imag;
        double i = this.real * c.imag + this.imag * c.real;
        return new Complex(r, i);
    }

    Complex divide(Complex c) {
        double denom = c.real * c.real + c.imag * c.imag;
        double r = (this.real * c.real + this.imag * c.imag) / denom;
        double i = (this.imag * c.real - this.real * c.imag) / denom;
        return new Complex(r, i);
    }

    void display() {
        System.out.println(this.real + (this.imag >= 0 ? " + " : " - ") + Math.abs(this.imag) + "i");
    }
}

public class ComplexNos {
    public static void main(String[] args) {
        Complex c1 = new Complex(3, 2);
        Complex c2 = new Complex(1, 7);

        c1.display();
        c2.display();

        c1.add(c2).display();
        c1.subtract(c2).display();
        c1.multiply(c2).display();
        c1.divide(c2).display();
    }
}
```

Var 5 :



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```
class Complex {
    double real, imag;

    Complex(double real, double imag) {
        this.real = real;
        this.imag = imag;
    }

    Complex add(Complex c) {
        return new Complex(this.real + c.real, this.imag + c.imag);
    }

    Complex subtract(Complex c) {
        return new Complex(this.real - c.real, this.imag - c.imag);
    }

    Complex multiply(Complex c) {
        double r = this.real * c.real - this.imag * c.imag;
        double i = this.real * c.imag + this.imag * c.real;
        return new Complex(r, i);
    }

    Complex divide(Complex c) {
        double denom = c.real * c.real + c.imag * c.imag;
        double r = (this.real * c.real + this.imag * c.imag) / denom;
        double i = (this.imag * c.real - this.real * c.imag) / denom;
        return new Complex(r, i);
    }

    void display() {
        System.out.println(this.real + (this.imag >= 0 ? " + " : " - ") + Math.abs(this.imag) + "i");
    }
}

class ComplexOperations {
    static Complex conjugate(Complex c) {
        return new Complex(c.real, -c.imag);
    }

    static Complex inverse(Complex c) {
        double denom = c.real * c.real + c.imag * c.imag;
        return new Complex(c.real / denom, -c.imag / denom);
    }

    static void polarForm(Complex c) {
        double r = Math.sqrt(c.real * c.real + c.imag * c.imag);
        double theta = Math.atan2(c.imag, c.real);
        System.out.println("Polar Form: " + r + " (cos(" + theta + ") + i sin(" + theta + "))");
    }
}

public class ComplexNos {
    public static void main(String[] args) {
        Complex c1 = new Complex(3, 2);

        System.out.print("c1 = "); c1.display();

        System.out.print("Conjugate = "); ComplexOperations.conjugate(c1).display();
        System.out.print("Inverse = "); ComplexOperations.inverse(c1).display();
        ComplexOperations.polarForm(c1);
    }
}
```




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Post Lab Subjective/Objective type Questions:

1. Understand the following code and rewrite the code as per inform in following question.

```
class SalesTaxCalculator {  
    float amount=100.0f; // instance variable  
    float taxRate=10.2f; //instance variable  
    void calculateTax() {  
        float taxAmt = amount*taxRate/100;  
        System.out.println(taxAmt); }  
    public static void main (String args[ ]) {  
        SalesTaxCalculator obj1 = new SalesTaxCalculator();  
        SalesTaxCalculator obj2 = new SalesTaxCalculator();  
        System.out.println("Amount in Object 1: "+ obj1.amount);  
        System.out.println("Tax Rate in Object 1: "+ obj1.taxRate);  
        System.out.println("Amount in Object 2: "+ obj2.amount);  
        System.out.println("Tax Rate in Object 2: "+ obj2.taxRate);  
    }  
}
```

- a) Modify to accept instance variable using a constructor with no argument and execute it.
 - b) Overload the constructor in part (a) and execute it.
 - c) Make the use of **this** keyword in part (b) and show its usages.
2. Explain the difference between instance variables and class variables.
 3. Explain the different types of constructors.

1. a)



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```
class SalesTaxCalculator {  
    float amount;  
    float taxRate;  
    SalesTaxCalculator() {  
        amount = 100.0f;  
        taxRate = 10.2f;  
    }  
    void calculateTax() {  
        float taxAmt = amount * taxRate / 100;  
        System.out.println("Tax Amount: " + taxAmt);  
    }  
    public static void main(String[] args) {  
        SalesTaxCalculator obj1 = new SalesTaxCalculator();  
        obj1.calculateTax();  
    }  
}
```

1. b)



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```
class SalesTaxCalculator {  
    float amount;  
    float taxRate;  
    SalesTaxCalculator() {  
        amount = 100.0f;  
        taxRate = 10.2f;  
    }  
    SalesTaxCalculator(float a, float t) {  
        amount = a;  
        taxRate = t;  
    }  
    void calculateTax() {  
        float taxAmt = amount * taxRate / 100;  
        System.out.println("Tax Amount: " + taxAmt);  
    }  
    public static void main(String[] args) {  
        SalesTaxCalculator obj1 = new SalesTaxCalculator();  
        SalesTaxCalculator obj2 = new SalesTaxCalculator(200.0f, 5.0f);  
        System.out.println("Object 1:");  
        obj1.calculateTax();  
        System.out.println("Object 2:");  
        obj2.calculateTax();  
    }  
}
```



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1. c)



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```
class SalesTaxCalculator {  
    float amount;  
    float taxRate;  
    SalesTaxCalculator() {  
        amount = 100.0f;  
        taxRate = 10.2f;  
    }  
    SalesTaxCalculator(float amount, float taxRate) {  
        this.amount = amount;  
        this.taxRate = taxRate;  
    }  
    void calculateTax() {  
        float taxAmt = amount * taxRate / 100;  
        System.out.println("Tax Amount: " + taxAmt);  
    }  
    public static void main(String[] args) {  
        SalesTaxCalculator obj1 = new SalesTaxCalculator();  
        SalesTaxCalculator obj2 = new SalesTaxCalculator(200.0f, 5.0f);  
        System.out.println("Object 1:");  
        obj1.calculateTax();  
        System.out.println("Object 2:");  
        obj2.calculateTax();  
    }  
}
```



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2. Declared inside a class but can not be modified in a method, it is class specific and can not be modified inside a method,(instance variables are not static while class variables are static).

3. NoArgsConstructor, AllArgsConstructor or Parameterized constructors.(Default is NoArgsConstructor)

Conclusion:

Learned how to implement this keyword, different types of constructors, constructor overloading, static variables and instance variables, uses of constructors helps us define object and if the need parameters to initialize.

Signature of faculty in-charge with Date: